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sustAinability Project Report

Challenge

Activating AI-Driven Digital Democracy: How Can Civic Engagement Be Increased on Nexus Politics?

Team & Project Nexus Polis

Appealing Visual



1. Introduction

Our challenge as the Nexus Politics team started from a pattern we kept seeing in civic participation platforms: they tend to fail at the same three points. Acquisition: how do you reach citizens at all without a marketing budget behind you. Activation: how do you get someone who has just landed on the platform to actually do something meaningful within the first few minutes, instead of bouncing off. Retention: how do you turn that first action into something people come back to, rather than a one-off. These three are not separate problems; they build on each other. There is no point activating a user you never reached, and no point designing for retention if users never got activated in the first place. So the problem we were really working on was not a missing feature, but the whole journey a citizen goes through from the moment they open the platform. We approached this by building Nexus Politics as an interactive civic platform that connects German citizens (and residents) to local democratic issues, polls, and discussions via location-based mapping.

We built a clickable front-end for Nexus Politics that shows what the platform feels like to actually use. The homepage opens on an interactive map of Germany where you pick a state, a district, or your own municipality and see what people nearby are discussing. From there it splits two ways: vote in live polls, or start something yourself. That second path lets a citizen post a concern and, if they want, turn it into a poll, a survey, or a feedback round, then tag it to a location so it reaches the right people. There is an events feed and a personal profile that tracks what you have contributed. Everything runs in the browser with no login needed to explore. It is a demo, but it is designed to feel real.

Mapping this back onto the challenge, each part of the prototype targets one of the three failure points. The map-first homepage is our answer to activation: instead of landing on an empty dashboard, a user is dropped straight into something local and relevant within seconds. The "start something yourself" path, turning a concern into a poll, survey, or feedback round and tagging it to a location, is what we hope drives organic acquisition, since a citizen who posts something local has a reason to share it with people around them rather than the platform needing to advertise itself. The events feed and personal profile are where retention lives, giving participation a visible, accumulating trace instead of letting it disappear after a single click. The rest of this report goes through these pieces in more detail and explains the design decisions behind each one.

2. Problem Analysis

What Problem Are We Solving?

As described in the previous chapter, the problem consists of three layers: Acquisition, Activation, and Retention of platform users. These layers do not stand alone, though they build on each other. A platform cannot retain users it never activated in the first place, and it cannot activate users it never reached. So if a solution only fixes one layer, for example, building a great participation tool without also solving distribution, the underlying democratic deficit stays exactly where it was.

But this funnel is really just the surface. Underneath it lies a deeper, more structural problem: citizens feel disconnected from democratic institutions not because they don't care, but because they don't believe their voice makes a difference. Our own survey (50+ respondents, 85.8% aged 18–34) supports this: interest in public issues is clearly not the bottleneck, since roughly half of respondents reported high interest in politics and public affairs.

What breaks down is the link between caring and acting: 79% of respondents said they perceive their influence on political outcomes as low, and participation was consistently described as too slow, too unsafe, or too ineffective to be worth the effort. When asked what would make them participate, respondents converged on three conditions current civic infrastructure does not meet the participation needs to feel relevant (54.2%), visible (54.2%), and easy (37.5%). On top of this engagement gap sits a distribution gap: 93.8% of respondents had never heard of Nexus Politics at all, meaning the product-level problem (engagement design) cannot even be tested at scale until the awareness problem is solved.

These numbers are supported by broader statistics. The Konrad-Adenauer-Stiftung's analysis of the 2025 Bundestag election confirms that voters under 30 have below-average turnout rates and account for only 13% of all eligible voters: the group with the most pronounced volatility and the lowest institutional trust (KAS, 2026). The OECD's Government at a Glance 2025 further reports that young people aged 18–29 are 21 percentage points less likely to vote in national elections than those aged 50 and over, while simultaneously being more active than older generations on social media and in public demonstrations, confirming that the issue is not apathy but a structural disconnect between informal civic energy and formal participation channels (OECD, 2025).

Why Is This Challenge Complex?

There are some several factors making this challenge complex: The zero-budget constraint is what makes this genuinely difficult. Most growth playbooks assume some level of paid acquisition: social ads, search campaigns, influencer deals. None of that is on the table here, either by challenge design or by the simple reality that most civic-tech ventures do not have marketing budgets. That means the team has to rely entirely on organic mechanics: word-of-mouth, referral loops, community partnerships. These can work, but they are slower to build, harder to control, and offer no guaranteed return on effort.

The GDPR and neutrality constraints limit the growth-hacking toolkit. Many of the most effective engagement mechanics in consumer social apps (dark patterns, algorithmic outrage amplification, manipulative notifications) are explicitly disallowed here, since the brief requires GDPR compliance, political neutrality, and non-exploitative gamification. Growth must be achieved without the very mechanics that made comparable platforms (e.g., entertainment or social apps) viral in the first place.

Trust is a precondition, not a byproduct. Unlike a shopping or entertainment app, a civic platform's core product is trust: in data handling, in political neutrality, and in the idea that a submitted concern leads somewhere. Trust is slow to build and easy to lose, which conflicts with growth strategies that optimize for speed.

The target behavior is inherently low-frequency. Voting, petitioning, and civic deliberation are not naturally daily habits the way messaging or entertainment consumption are. Designing

"habit formation" for a low-frequency, high-effort behavior is a harder behavioral-design problem than habit-forming a low-effort, high-frequency one.

Why Hasn't It Been Solved Before?

Civic-tech and e-participation tools are not a new category — the European Parliament's 2026 STOA study reviewed 94 digital participation tools worldwide and found that well-designed platforms do improve the responsiveness of democratic processes (Bådum et al., 2026). Yet scale remains rare for recurring structural reasons. Most civic platforms are built institution-first: commissioned by municipalities to fulfill transparency mandates rather than designed around the Acquire–Activate–Retain discipline that consumer products rely on. Psychological frameworks such as Self-Determination Theory (Deci & Ryan, 1985) and nudge theory (Thaler & Sunstein, 2008) have proven effective at driving habitual behavior in commercial products, but their application to civic platforms remains limited (Hofmann & Pappas, 2021). This is partly deliberate: research on AI-mediated participation, including ongoing work at the TUM Institute for Ethics in AI, warns that the same behavioral techniques that increase engagement can if applied carelessly shift from empowering citizens to manipulating them (TUM IEAI, 2026; Ruehle, 2023). For a civic platform like Nexus Politics, where political neutrality and user trust are non-negotiable, this tension makes aggressive behavioral design a genuinely risky path.

User groups (Mana)

Because democratic participation affects society as a whole, the potential user group of Nexus Politics is very broad. In principle, all citizens and residents who are affected by political decisions could benefit from a platform that makes participation more accessible and visible. However, our project focuses especially on Gen Z and young, digitally active people, such as students and young professionals. This group is already familiar with digital platforms, but often does not participate regularly in formal political processes. For Nexus Politics, they are therefore a strategically important target group: they are reachable through digital and community-based channels, but they need low-friction, relevant, and trustworthy ways to engage.

Stakeholder Mapping (Mana)

The main stakeholders of Nexus Politics include citizens and platform users, the Nexus Politics team, politicians, municipalities, public institutions, civil society organizations, universities, student groups, and political education actors. Citizens are the primary users because they submit, support, discuss, and follow concerns. Politicians and municipalities are important because they can respond to concerns and connect digital participation to real decision-making processes. NGOs, universities, and community groups can act as trusted bridges between the platform and potential users. Finally, the Nexus Politics team is responsible for designing, maintaining, and moderating the platform in a way that remains neutral, transparent, and trustworthy.

3. Your Solution

Adan: Nexus Politics addresses the participation gap not by adding another top-down consultation channel, but by building a single infrastructure that connects citizens, politicians, and administrations around concrete local concerns. Citizens use the platform for free to submit concerns, follow legislation, respond to targeted polls, and locate the representatives responsible for the issues they care about; politicians and municipalities pay for a CRM, analytics, and survey layer that structures this input and makes their responses visible (Nexus Politics, n.d.). The result is a two-sided model in which the accountability mechanic, namely public response rates and traceable positions, is not a bolted-on feature but the core value exchange. This information matters because the participation deficit is not evenly distributed. Class background still strongly patterns who engages, who feels efficacious, and who trusts democratic institutions across Europe (Grasso & Giugni, 2022), so a solution that merely rewards the already-active would deepen the very inequality it claims to solve.

Our contribution builds on this foundation through gamification, but a deliberately cautious form of it. The literature is a warning as much as an invitation. Reward-based mechanics such as points, leaderboards, and missions repeatedly fail to activate the disengaged and mostly intensify participation among those already interested (Thiel et al., 2017). Worse, competitive game layers can generate friction that advantages networked, knowledgeable users and quietly excludes the rest (Tseng, 2023). Even where practitioners are open to gamification, they lack the guidelines, resources, and know-how to implement it well (Tóbiás & Boros, 2025). We take these findings as design constraints. Rather than a competitive points economy, our system weights civic actions by effort and impact, uses streaks with grace days to avoid punishing intermittent users, and nests any ranking within local pools so that participation is framed as a contribution to one's community rather than a contest between citizens.

The distinction between the old and new prototypes reflects this reasoning. The earlier version was a five-page clickable demonstration that established the visual language and core flows; the current homepage reorganises the product around what the live platform actually does, presenting an interactive map of Germany that layers concerns, events, polls, politicians, and elections, which can be filtered by region and category. Where the old prototype showed features, the new one shows relationships: a citizen tapping a state sees local activity, and the gamified profile makes their own accumulated impact legible without exposing them to a discouraging leaderboard. Users learn about the platform through the municipalities and representatives already using it, embedding adoption in existing civic structures rather than relying on paid marketing. Because the whole system is GDPR-compliant and hosted in Germany, it can carry this participation data credibly in a low-trust environment, a precondition rather than a detail, given that distrust is among the strongest barriers to engagement (Tóbiás & Boros, 2025).

Diwen: Therefore, this trust will consist of two aspects: trust between people and self-confidence. According to our survey, many people are dissatisfied with the current socio-political climate, or even uninterested, because they feel their voices are insignificant or

even ignored. The survey also revealed that many people, especially young people, consider themselves politically marginalised, thus lacking both of these types of trust. Unlike existing civic participation platforms that mainly collect feedback, our solution integrates gamification, public accountability, and geographically localised participation into a single ecosystem. Our improved platform can change this through our gamification features. Our gamification features are designed based on the principle of "all participation counts." All forms of participation, from attending offline rallies to checking in or voting, will be rewarded with "levels and experience", allowing participants to cultivate a habit of political participation in a short period and receive feedback quickly, which could build the "political confidence" for users. Furthermore, we have added "Vote for Yourself" and "Voice yourself" on the homepage to encourage users to participate in political discussions. Interactive maps allow users to participate in rallies in the same or different regions, which encourages participation in "political life." Through these online and offline methods, trust between people will be built, which enables users who find their echo chambers and encourages them to embrace different voices. If this approach can be promoted through Nexus Polis, democracy will be further strengthened, and social participants will engage in politics in a more direct and bottom-up manner, new social forces in the real world may be formed. Of course, the platform should bear relevant social responsibilities, such as preventing addiction and stopping the spread of illegal information (e.g., racial discrimination, support for Nazi ideology). Anti-addiction mechanisms and user behaviour guidelines will be communicated to users before they use the platform, and the platform will adhere to the principles of transparency and timeliness, while respecting user privacy and maintaining openness and transparency of information in accordance with the EU GDPR and other laws. However, based on the principles of democracy and trust, Nexus Polis will not introduce any proactive censorship. If any abuse of the platform or attempt to spread illegal information is discovered, users will become the adjudicators, inducing "passive adjudication" through reporting, voting, or red flags. Whether illegal information is in violation will be determined by the majority of users. Users will become the main actors in the platform's operation. In the future, our platform will continue to incorporate more diverse ways of political participation and become a bridge between politicians and the public. Users may have the opportunity to directly launch petitions on the platform or participate in politics in a more direct way in a similar manner. This barrier-free participation is the ideal way to promote trust between the two groups, which is also essential in political participation in the real world. Building on this foundation, we will also launch "For Groups" and "For Organizations" versions, allowing organizations and groups playing different social roles to participate in political life in a more direct and transparent manner. Here, not only can organizations and groups participate directly in real political life on a representative basis, but they can also use this platform to promote transparency and democracy within their organizations. Therefore, all users aged 18 and above are encouraged to participate.

The implementation of our solution will be a key focus. The municipal government will subscribe to the platform, and all municipal staff will have access to participate. Simultaneously, we will reject all subscriptions made in the name of political parties. Politicians without direct positions in the municipal government can participate individually or as politicians, and will be required to indicate their party affiliation and any parties they have previously represented. Ordinary citizens can participate individually for free. Group participation in the municipal

government version is free; groups also need an internal version (limited to group members) which can be accessed through subscriptions or purchases. When users initiate opinion polls, votes, complaints, or other forms of feedback, the opinions will be uploaded to the cloud, visible and accessible to users in the user's designated area. This content will be continuously and repeatedly pushed to trending topics based on AI algorithms. Once the number of views or participation reaches a certain level, the content will be featured on the regional popularity chart, and politicians will be notified. A Customer Relationship Management system will be used to organise, categorise, and distribute citizen-submitted questions. For example, questions related to "transportation," "housing," and "education" will be automatically categorised and assigned to the appropriate councilors or departments. The system analyses the received data, such as: which issues are of most concern? Which regions have the most reported problems? Which councilors respond the fastest? It then generates reports for government or political figures to review. Simultaneously, if a particular issue requires more information, the platform sends questionnaires to specific regions or groups. Politicians or government departments respond publicly on the platform, and citizens can see the responses and processing progress. After users complete participation (e.g., submitting issues, voting, filling out questionnaires, participating in activities), the system automatically increases experience points (XP), levels, or badges. If a petition is initiated, the platform will replace the original municipal platform as a "Gate Keeper", completely avoiding the situation where the municipal government "acts as both player and referee." Once a petition passes and is released to the platform's public area, the platform will collect user feedback and signatures. Once a certain number is reached, the relevant political users are obligated to make a public response within a certain timeframe. Of course, other inspirations from users' feedback will be further considered.

4. Discussion & Outlook

Value Proposition (Sean)

As described in previous chapters, our research showed that low voting turnout among young voters and low awareness of platforms built to change that remain as major issues in today's democracy. Nexus Politics addresses the gap with the utility of the product, in an attempt to motivate users to direct participation.

At its core sits a live interactive map where users can see what their region is debating in real time and explore local issues before posting. Around it, citizens can vote in real polls, answer quick surveys, and watch regional hotspots light up across the country, so that participation feels current and grounded in place rather than abstract. Events and communities let users discover nearby movements, join the ones that matter, and organise with others, while a clean personal profile shows their progress, history, and reach. Together these features turn a vague sense of civic duty into something visible, local, and easy to act on. The roadmap deepens this value over time. Petitions and dialogues will run through a third party bridge that keeps every process transparent and makes direct exchange with politicians possible. Internal versions will bring the platform to associations, universities, and other organisations at different scales, with dedicated chatrooms. Further gamification will widen the funnel to new participants.

Value also depends on reach, and the go-to-market approach is built to grow without a large budget. Digital channels such as quests, digital badges, and social campaigns run alongside traditional formats like student ambassadors, city quests, school visits, and QR handouts in public places. This is reinforced by partnerships with civic participation platforms including *Mehr Demokratie e.V.*, *Initiative Offene Gesellschaft*, and *Abgeordnetenwatch*. A particularly natural fit is *Beteiligungs-KI*, an AI tool that *Initiative Offene Gesellschaft* and *Mehr Demokratie e.V.* are developing to help municipalities design participation processes, where Nexus Politics can supply the real citizen feedback layer that complements its planning function (Initiative Offene Gesellschaft, n.d.). Alongside these sit political education institutions such as the *bpb*, *Die Junge Akademie*, and *Demokratie leben!*, and with referral programs across universities and other businesses.

Flaws and Risks (Sean & Mana)

Although our solution introduces an optimistic future for democracy, it also carries risks that deserve honest treatment. The most immediate concern is gamification itself. Reward mechanics designed to build civic habits sit close to the nudging patterns that sustainability assessments of AI systems warn against (Rohde et al., 2024), and the line between motivating participation and manipulating it is thinner than a feature list suggests. Our tiers therefore reward only genuine civic acts rather than time on platform, but this boundary requires ongoing scrutiny rather than a one-time design decision. A second risk lies in the intelligence layer. AI that clusters and prioritizes citizen concerns can quietly shape which voices count, and the same capabilities that structure political input at scale have been shown to carry potential for bias and opinion steering (Vinuesa et al., 2020). Transparency about how concerns are processed is thus a democratic requirement, not a compliance detail. Finally, a platform-first model risks reducing participation to data collection while amplifying the already digitally active and leaving quieter groups behind (Steig et al., 2025). Partnerships with offline participation formats and accessible design are our hedge against this, yet the risk cannot be engineered away entirely and should be monitored as the platform scales.

Another possible flaw of the Nexus Politics concept is the risk of creating an expectation gap. Our survey showed that users are strongly motivated by visible impact, updates, and measurable results after participation. This makes an impact tracker and transparent feedback loop highly valuable. However, Nexus Politics cannot fully control whether municipalities, politicians, or institutions actually respond to citizens' concerns in a timely and meaningful way. If users submit or support a concern but later see no real progress, the platform may unintentionally reinforce the same frustration it tries to solve: the feeling that civic participation does not matter. In this case, the problem would not only be low engagement, but disappointed engagement.

Next Steps (Mana)

A first important step would be to further develop and test the redesigned user experience presented in our prototype. The current Nexus platform already offers civic participation functions, but the proposed redesign makes the entry point more visual, local, and user-friendly. Features such as the interactive map, topic filters, polls, events, and user profile should be

refined into one coherent user journey. A first-time visitor should quickly understand what is happening around them, why a topic is relevant, and what simple action they can take next. This is important because the platform should not feel like a complex political tool, but like an accessible entry point into civic participation.

The redesign should also focus on connecting individual features more clearly. The map should help users discover local concerns, polls should offer a low-effort way to participate, events should connect online activity with real-world communities, and the profile should make a user's participation history visible. These elements should not feel like separate sections, but like connected steps in one participation process. In this way, Nexus can support users who only want to explore at first, while still creating a path toward deeper engagement later.

Plus, Nexus Politics should evaluate future growth through more than simple visibility metrics. It is not enough to know how many people see a campaign or visit the website. More useful indicators would be how many users complete a first action, how many return, which topics generate repeated engagement, and which communication channels bring users who actually participate. This would help Nexus improve step by step and focus on strategies that support long-term civic engagement rather than short-term attention.

Before scaling further, Nexus should also define internal decision rules for feature development. For example, every new feature should answer one of three questions: Does it help users discover relevant issues? Does it make participation easier? Does it make the outcome of participation more visible? This would prevent the platform from becoming too complex too early and would keep the next development phase focused on the original goal of increasing civic engagement.

In the long term, Nexus Politics can become a more familiar and trusted platform for digital democratic participation. To reach that point, the next phase should combine the existing platform with the proposed redesign, focused communication, meaningful follow-up, and strong partnerships. This would turn Nexus from a platform that people may visit once into a civic tool that people understand, use, and return to.

5. Author Contribution

1. *Introduction* — Melike Sert
2. *Problem Analysis* — Amina Khussainova & Mana Adibi
3. *Your Solution* — Diwen Zheng & Adan Abdulrahman
4. *Discussion & Outlook* — Yu-Hsiang (Sean) Chen & Mana Adibi

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